

High Content Optical Mapping for Cardiac Electrophysiology

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Research Summary

Cardiac electrophysiology depends on understanding how electrical impulses are generated and propagate across heart tissue, which is central to diagnosing and treating arrhythmias. Optical mapping is a powerful experimental technique that uses a transmembrane voltage-sensitive fluorescent dye together with a high-speed image sensor to visualize action potential propagation across the surface of the heart at high spatial and temporal resolution, recording activity from many locations simultaneously so that the shape of action potentials and the spread of activation and repolarization can be observed directly. The method is, however, extremely data- and compute-intensive: analyzing even a few seconds of recorded imaging can require hours of processing on conventional hardware, which substantially limits its experimental utility. This work investigates a high-content optical mapping pipeline and the hardware- and software-acceleration strategies needed to make such analysis tractable, and ultimately real-time, several orders of magnitude faster than the state of the art at the time. Making optical mapping fast enough to run in real time opens new avenues of research, including closed-loop feedback control of cardiac function that is not feasible using coarse global measures of electrophysiology such as the electrocardiogram (ECG).

Brief research summary prepared for reference. The original was presented as a workshop paper at the International Workshop on Biomedical System Design (BSD 2010) and was not formally published; no archival/DOI version exists.